Recovering Content Performance Archetypes with Validated K-Means Clustering

Tracy Zhang

Abstract

What recurring performance archetypes characterize a multi-client search engine optimization content inventory, and can these archetypes be recovered by an unsupervised method without recourse to hand-specified thresholds? This study analyzes 107,523 pages drawn from one month of the FlyRank AI internship data on Huggingface across 50 clients, characterizing each page by click-through rate, ranking position, content staleness, impression volume, and on-page engagement rate. K-Means clustering is applied to these five behavioral signals, with the number of clusters selected by silhouette score and the resulting model validated through a client-grouped train and test split, an independently constructed rule-based baseline, and a leakage attack test. The selected model, comprising 20 clusters, recovers five of seven named archetypes and surpasses both the rule-based baseline and a random-label floor on every internal quality metric, while exhibiting only limited agreement with that baseline, reflected in an Adjusted Rand Index of 0.098. The resulting output is a ranked action list assigning each archetype to a corresponding recommendation to protect, improve, rewrite, merge, prune, or monitor.

1 Introduction

Content operations teams managing many clients cannot review search and analytics metrics for every page manually. A rule-based system was built to help, sorting pages into seven named archetypes, namely champions, hidden gems, rising stars, stale visible pages, weak or no-demand pages, engagement problem pages, and cannibalization risk, on the basis of explicit thresholds applied to position, volume, click-through rate, staleness, and engagement. The claim underlying that system is that these seven hand-specified categories correspond to the actual recurring patterns in how content performs and that a fixed set of thresholds is sufficient to sort any page into the correct one. Its thresholds were set once without subsequent reassessment and by construction, it cannot surface a pattern it was not built to anticipate. This paper checks whether that claim holds against the underlying structure of the data.

This study investigates that claim directly: whether an unsupervised method, K-Means clustering applied to the same underlying behavioral signals, can recover recurring page types independently and where the resulting groupings converge with or diverge from the hand-written rules. The present analysis is intended to inform prioritization decisions, specifically which pages within a large content inventory warrant protection, improvement, rewriting, merging, pruning, or monitoring.

2 Data

The data for this study are drawn from the FlyRank AI internship data on Huggingface, covering every `report_date` recorded in the fact table between June 1 and June 30, 2026. Content staleness is calculated relative to the fact table's own maximum report date rather than the current calendar date, ensuring that the staleness measure cannot reference a period beyond the data's coverage.

Joining aggregated search and analytics metrics to published, non-deleted content yielded a total of 394,928 page-level rows. Of these, 272,549 rows, corresponding to 69% of the total, were excluded due to the absence of usable Google Search Console data or fewer than 10 total impressions, as a rule or cluster derived from near-zero signal cannot be considered reliable. This exclusion criterion operates on the same impression and availability fields from which the clustering features are subsequently derived, and therefore does not constitute a neutral cut: pages with insufficient measurable activity are never given the opportunity to be clustered or assigned an archetype, a scope limitation disclosed explicitly rather than left implicit. The remaining 122,379 rows constitute the modeling population, partitioned by client into 107,523 training rows and 14,856 test rows, as detailed in the Methodology section.

No article text, keyword text, or personally identifying fields were used at any stage of this work. Every client and content identifier is represented as a hashed value rather than a name or a URL, and no identifying text of this kind appears anywhere in this paper.

3 Methodology

3.1 Baseline

A rule-based archetype label was constructed prior to and independently of any clustering procedure, using explicit thresholds applied to position, impression volume, click-through rate relative to a position-adjusted expectation, staleness, and engagement. Each page is assigned one of seven named archetypes, which are champions, hidden gems, rising stars, stale visible pages, weak or no-demand pages, engagement problem pages, or cannibalization risk, or otherwise a catch-all monitor category when no other condition is satisfied. This label constitutes the baseline against which the clustering model is subsequently evaluated.

In addition to the archetype label, each usable page receives a simple additive `action_score`, comprising points allocated for the magnitude of its position-adjusted click-through rate gap, for elevated traffic volume, for staleness, for diminished engagement, and for a confirmed cannibalization conflict. Each term is a fixed value determined by the directness of that signal's actionability. As none of these values are fit or learned, the resulting score remains fully auditable through direct inspection of the function that generates it.

The resulting ranking was evaluated against two weaker alternatives and the base rate. The proportion of usable pages carrying any substantive action, meaning any recommendation other than monitor, is 53.9%, representing the minimum threshold a ranking must exceed to justify its computation. A purely random ordering approaches this threshold at every depth examined, confirming that the evaluation procedure itself behaves as expected. Ranking by raw traffic volume alone, an approach an analyst might adopt in the absence of any rule-based system, achieves a precision between 70% and 85% within the top 20 to 1,000 pages. Ranking by `action_score`, by contrast, achieves a precision of 1.000 through the top 500 pages before declining to 0.624 by rank 1,000, as reported in Table 1.

This perfect precision at the highest ranks does not reflect an even distribution of scores: all 20 top-ranked pages share an identical value. Each of these pages ranks in position 1 to 3, receives real impressions ranging from approximately 120 to 670 with zero clicks and zero engaged sessions. A page ranking at the top of the first results page while accumulating genuine traffic but no clicks whatsoever constitutes an anomalous pattern, one consistent with a defective title or meta description, a search results feature intercepting the click, a branded query that does not require a click-through, or a measurement discrepancy unrelated to the page itself. `action_score` correctly identifies this subset of pages as the highest priority, though it cannot determine which explanation applies, a determination that requires the kind of manual review a practitioner must undertake before a rewrite is initiated.

3.2 Model

Five features are used for the final clustering model: click-through rate, average position, days since last update, a logarithmic transform of total impressions, and engagement rate, with missing engagement values imputed as zero, since a missing engagement rate specifically indicates that zero GA4 sessions were recorded rather than that engagement was inadequately measured.

The number of clusters, k, was determined through an exhaustive search across values from 3 to 30, with each value scored on the conventional elbow measure of inertia and on silhouette score, restricted to configurations in which no single cluster comprised more than 45% of rows. Silhouette score varies within a comparatively narrow band across most of this range; the silhouette-optimal balanced configuration occurred at $k = 8$, yielding a silhouette score of 0.360, whereas $k = 20$ was selected explicitly, sacrificing a modest degree of separation, 0.327 relative to 0.360, in exchange for a model with sufficient clusters to recover five of seven archetypes rather than merging them together. A per-cluster naming confidence indicator relative to the rule-based baseline is reported, but this measure does not inform the present choice of k; silhouette score alone determines it, ensuring that the model is not implicitly optimized toward agreement with the very rules it is intended to evaluate independently.

3.3 Validation Design

The training and test sets were split by client rather than by individual page. Because all pages belonging to a given client remain entirely within a single partition, no client-level pattern, such as a characteristic update cadence or a tracking irregularity, can leak across the split and artificially inflate agreement between training and test performance. This design choice was verified directly by fitting the identical final configuration under a naive random row-level split and comparing client overlap and silhouette score between the two approaches, with results reported in the Results section.

3.4 Leakage Audit

Standard diagnostic checks were conducted against the final feature set and split. The staleness reference date remains within the fact table's own temporal window, no rows exhibit a negative days-since-update value, no product flag or prior-system decision, such as `provider_used`, `model_used`, or `optimization_eligible_date`, appears among the clustering inputs, and neither the archetype label, the cannibalization flag, nor the keyword identifier leaked into the feature set.

Passing these diagnostic checks does not, by itself, demonstrate that the audit would successfully detect an actual leak; the audit procedure was therefore deliberately subjected to an adversarial test. `ctr_gap`, the position-adjusted click-through rate gap on which several archetype rules threshold directly, was introduced into the feature set, and the model was refit at the same $k = 20$; agreement with the rule-based archetype shifted only marginally, with the Adjusted Rand Index moving from 0.098 to 0.110. Interpreted at face value, this result might suggest that the audit lacks sensitivity to leakage, prompting a second and more direct injection: a label-encoded representation of the archetype itself, information to which the honest model has no access. This second injection produced a substantial increase, to 0.357, more than three times the original score, resulting from the addition of a single column. The initial test proved weak not because the audit fails to detect genuine leakage, but because `ctr_gap` closely approximates a fixed transformation of two features already present in the model, namely click-through rate and position. The second test confirms that the audit does, in fact, detect such leakage when it is genuinely present.

4 Results

4.1 The Chosen Model

The final model achieves a training silhouette score of 0.327, compared with 0.302 on the test set, across cluster sizes ranging from 22 to 16,942 rows. Of the 107,523 training pages, the model successfully names five of the seven named archetypes above a 35% naming confidence threshold: champions comprising 8,181 pages, hidden gems comprising 1,253 pages, rising stars comprising 18,581 pages, engagement problem pages comprising 11,536 pages, and weak or no-demand pages comprising 20,462 pages. Stale visible pages was not recovered under any tested configuration, as only 27 rows carry this rule-based label across the entire 394,928-row dataset, an insufficient quantity for any cluster to concentrate around. The remaining 47,510 pages, representing 44.2% of the training set, fall into clusters below the naming confidence threshold and are accordingly labeled needs review rather than assigned to a weak match.

4.2 Archetypes and Their Recommended Actions

The archetypes named above, together with cannibalization risk, located directly from the underlying data rather than through the clustering procedure, jointly define a primary output of this study: a mapping from each archetype to a single recommended action. This mapping was specified in advance, as part of the rule-based baseline described in the Methodology section, rather than fitted

Table 1: Precision at K: share of the top K ranked pages the rule engine itself calls actionable, against two weaker rankings and the base rate.

K	action_score	traffic only	random shuffle	base rate
20	1.000	0.700	0.650	0.539
50	1.000	0.760	0.600	0.539
100	1.000	0.830	0.490	0.539
500	1.000	0.844	0.528	0.539
1000	0.624	0.850	0.539	0.539

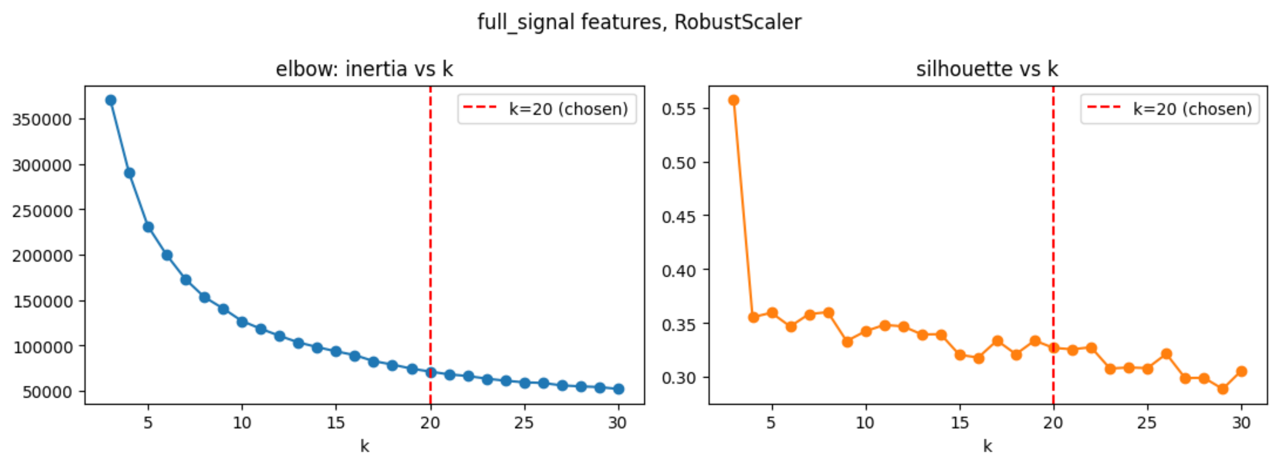

Figure 1: Inertia and silhouette score plotted against k for the five-feature model with a robust scaler. Neither metric exhibits a single unambiguous inflection point; silhouette score varies within a narrow band across most of the range, which is why the selection of $k = 20$ required an explicit justification rather than an automated selection procedure.

after the fact to the composition of any particular cluster, so that the action attached to a page follows from its measured behavior rather than from a judgment made once the clusters were already in hand. Table 2 reports the resulting action and page volume for each named archetype, together with the needs review category, within the training set. This mapping is a category-level signal rather than a page-specific determination, it applies only to the 55.8% of training pages carrying a confident archetype, and it should be read together with the caveats set out in the Limitations section, particularly regarding which actions must never be automated and the plurality basis of naming confidence.

Each action follows from a specific pattern among the five underlying signals rather than from the archetype's label alone. Champions combine a strong position with a click-through rate that meets or exceeds expectation, warranting protection rather than intervention. Hidden gems pair that same quality with comparatively low volume, warranting improvement in the form of promotion rather than a rewrite. Engagement problem pages combine strong visibility with weak on-page engagement, a mismatch that rewriting is intended to correct. Weak or no-demand pages score poorly across position, click-through rate, and volume alike, leaving pruning as the only action likely to be worth the effort involved. Rising stars already exceed expectation on a favorable trajectory, warranting continued monitoring rather than immediate action. Cannibalization risk was located directly from two pages that share both a client and a keyword rather than inferred from a cluster, and merging is warranted once a person has read both pages. Needs review, comprising every page whose cluster falls below the 35% naming confidence threshold, receives no automated action and is instead directed to monitoring through manual review. The practical considerations attached to each of these actions are addressed in the Recommendations section.

Exceeding a random floor represents the more elementary finding, as illustrated in Figure 3. Surpassing the rule-based baseline constitutes the more specific claim, and the model accomplishes this on silhouette score, 0.327 relative to 0.008, on the Davies-Bouldin index, 0.897 relative to 3.051, where lower values indicate better performance, and on the Calinski-Harabasz index, 64,753 relative to 6,046. Agreement between the two labeling schemes remains low, reflected in an Adjusted Rand Index of 0.098 and a Normalized Mutual Information of 0.285, indicating that the model is not merely reproducing the rule-based classifications. Rather, it identifies distinct structure within the same behavioral signals, which is why both a quality comparison and an agreement measure are reported jointly.

The closer agreement between training and test performance under the naive split, a gap of 0.005 relative to 0.025, does not

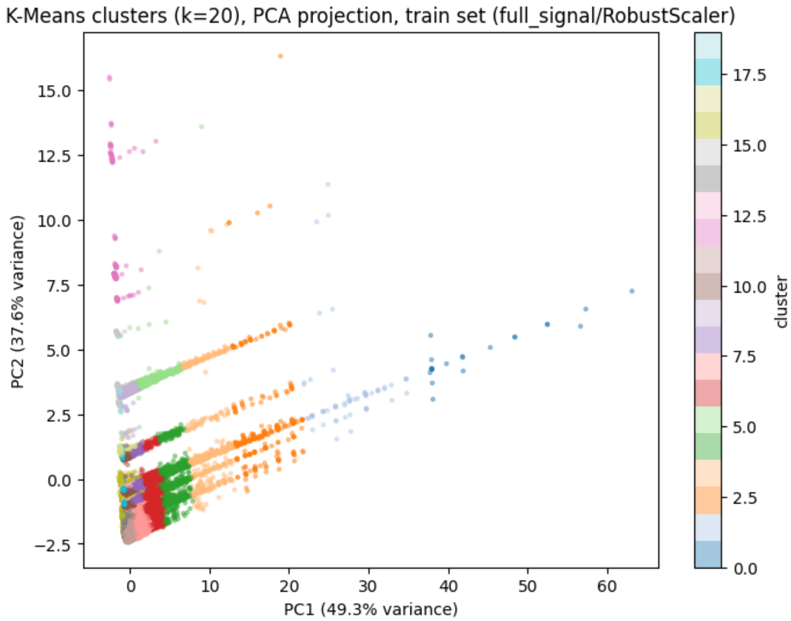

Figure 2: Twenty clusters projected onto their first two principal components, which jointly account for 86.9% of total variance. The clusters form visibly distinct regions rather than a single continuous mass, although several adjacent clusters share an indistinct boundary.

Table 2: Recommended action by archetype, training set.

Archetype	Pages	Action
Needs review	47,510	Monitor, manual review
Weak or no-demand pages	20,462	Prune
Rising stars	18,581	Monitor
Engagement problem pages	11,536	Rewrite
Champions	8,181	Protect
Hidden gems	1,253	Improve
Cannibalization risk	2	Merge

Table 3: Training and test silhouette scores under a naive random split compared with the client-grouped split actually employed.

Split type	Train sil.	Test sil.	Train-test gap	Client overlap
Naive random row	0.315	0.310	0.005	47 of 50
Grouped by client	0.327	0.302	0.025	0 of 50

constitute evidence of superior generalization; rather, it reflects the fact that 47 of 50 clients appear on both sides of that split. The grouped split's larger gap, accompanied by zero client overlap, represents the more accurate figure.

5 Limitations

This model constitutes decision support for prioritization. It is not a ranking algorithm, not an SEO score, and not a substitute for reading the pages it flags. Every claim advanced in this paper is observed, measured, directional, or offered as decision support, rather than established as proof of a causal or predictive effect. The rule-based baseline is itself a collection of hand-specified thresholds rather than ground truth regarding actual search outcomes, and consequently, agreement or disagreement with it characterizes structural similarity, not correctness.

Stale visible pages remains structurally unreachable given current data volume: the twenty-seven rows present in the full dataset are too few for any cluster to concentrate around, irrespective of feature set or k. The needs review category encompasses 44.2% of training rows; for this proportion, the model explicitly declines to

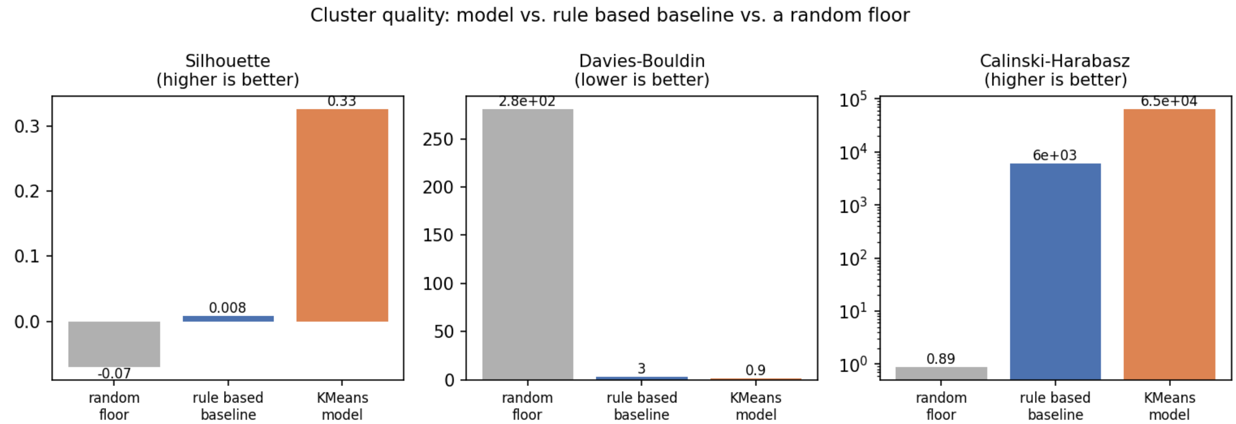

Figure 3: Silhouette, Davies-Bouldin, and Calinski-Harabasz values for a random 20-way label assignment, the rule-based archetype treated as a clustering solution, and the K-Means model, computed on identical scaled features. Both substantive methods exceed the random floor by a considerable margin on every metric, and the model additionally surpasses the rule-based baseline on all three.

assign an archetype rather than forcing an unreliable match, and these pages should accordingly be treated as unscored rather than presumed adequate.

Validation in this study is conducted against the rule-based baseline and a random floor, not against genuine outcomes such as ranking movement or conversion change. Low agreement with the baseline indicates that the two methods delineate different structural boundaries, not that either is confirmed to be correct. This model reflects a single data extraction covering June 2026 and a single client-grouped split; it has not been evaluated across multiple time windows, and a page's archetype derived from this snapshot is therefore not guaranteed to remain valid in subsequent months. No sealed or blind holdout is claimed anywhere in this project: the test split is reused openly throughout for comparative purposes, and test-side cluster labels are generated by applying the already-fitted model's prediction function, never through a second independent fit.

Five labeled clusters, comprising as few as 22 to 66 rows, report naming confidence percentages presented in the same format as clusters containing thousands of rows; although a small cluster's elevated confidence score is genuine, it rests on comparatively thinner evidentiary support. The composition of the modeling population is itself a methodological choice: excluding pages with low or no traffic prior to modeling means these pages are never afforded the opportunity to receive an archetype designation, a scope limitation disclosed explicitly rather than concealed. Relatedly, a cluster's naming confidence between 35% and 40% reflects a plurality rather than a majority, meaning most pages within that cluster do not necessarily match the assigned label, and any cluster whose composition appears dominated by a single client warrants a client level explanation, such as a tracking irregularity or a content operations habit, before being treated as an independent finding.

The archetype-to-action mapping reported in Table 2 is a category-level signal, applied uniformly across every page within a cluster, and certain actions built on it should never be automated on the basis of this model alone. Removing or unpublishing a page under the prune recommendation without human confirmation that it holds no non-search value is destructive and difficult to reverse. Merging or redirecting pages under the merge recommendation without a person reading both pages risks destroying a legitimately distinct page along with its backlinks and rankings. Any action taken on a needs review cluster beyond flagging it for human attention exceeds what this model claims, since no archetype determination is made for those pages by design. Reporting language should describe a cluster's designation as a plurality vote with an associated confidence score, not assert it as a certified classification.

Several concrete signals would indicate that these recommendations have grown stale and warrant reexamination: the average naming confidence across clusters falling below the 0.40 to 0.46 range observed throughout this project, the needs review share growing materially past its current 44.2%, a previously confident cluster such as champions crossing below the 35% naming confidence threshold, the Adjusted Rand Index or Normalized Mutual Information shifting noticeably from their current values, a change to the rule engine's own thresholds, a change to the underlying data pipeline, or the stale visible pages count growing meaningfully past its current 27 rows. Any of these developments would justify rerunning the clustering and validation procedure rather than continuing to rely on the present naming assignments.

6 Ranked Recommendations

Table 2 in the Results section reports the action and page volume associated with each archetype. The list below addresses each action in descending order of page volume, focusing on the practical caveats a practitioner should observe before acting.

Weak or no-demand pages. This constitutes the largest actionable category, comprising 20,462 pages for which pruning is recommended. Prior to removing any page, practitioners should confirm that it does not serve a non-search purpose and is not simply too recently published to have accumulated traffic.

Rising stars. This category of 18,581 pages exhibits genuine traffic volume alongside a click-through rate that exceeds expectations for its position, and monitoring is recommended. These pages warrant continued observation for potential promotion rather than immediate action.

Engagement problem pages. This category of 11,536 pages combines strong search visibility with diminished on-page engagement, and rewriting is recommended. Practitioners should first determine whether a given page functions as a quick-answer page before attributing low engagement to a content deficiency, prioritizing the highest-scored pages within this category for review. Several of the highest-ranked pages occupy position 1 to 3 with real traffic yet register zero clicks, warranting examination prior to initiating a rewrite.

Champions. This category of 8,181 pages represents the strongest cluster with respect to both position and click-through rate, and the recommended course of action is to preserve these pages as they stand while monitoring for regression.

Hidden gems. This category of 1,253 pages exhibits limited volume but substantial quality, and promotion is recommended rather than rewriting.

Cannibalization risk. These two pages were identified directly from two pages sharing a client and a keyword, rather than inferred from a cluster, rendering the confidence in this determination structural rather than statistical in nature. Both pages should be reviewed prior to merging.

Needs review. This constitutes the largest single category, comprising 47,510 pages for which no automatic archetype designation is assigned. A lighter-weight, lower-cost review process represents the appropriate next step, rather than automation.

7 Reproducibility

Repository. https://github.com/tracy030115/flyrank-ml/tree/main/submission/work/notebooks

8 Acknowledgments

This work is built upon the FlyRank ML Internship dataset.